WORKED SOLUTION

STEP III 2003, Question 13

Probability and Statistics • Absorbing Markov Chains

STEP III 2003, Question 13

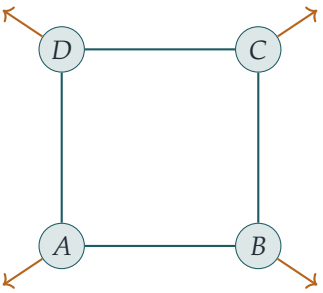
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q13

The key idea

This is a "first-step analysis": condition on what the rabbit does next, and the probability from one chamber is written in terms of the probabilities from the chambers it can reach. That gives four linear equations, and the symmetry of the square (reflecting in the diagonal AC swaps B and D) cuts the work down. For the final part we use the same trick on a modified warren in which entering C counts as failure, then apply the definition of conditional probability.

Solution



tunnels to the surface

Throughout, each move is chosen with probability $\frac{1}{3}$ independently of the rabbit's past (this is how we read "equally likely" for a rabbit "finding itself" in any chamber). Write E_A for the event that the rabbit eventually leaves through the tunnel from A.

Part (i)

Suppose the rabbit is in A, and condition on its next move.

- With probability $\frac{1}{3}$ it goes up the tunnel at A at once, and E_A has happened.

- With probability $\frac{1}{3}$ it runs to B. It is now in B, and because its future moves don't depend on how it got there, the probability that it eventually leaves from A is p_B.
- With probability $\frac{1}{3}$ it runs to D, and similarly the probability is then p_D.

By the law of total probability,

$$p_A = \tfrac{1}{3}\cdot 1 + \tfrac{1}{3}p_B + \tfrac{1}{3}p_D,$$

as required.

Part (ii)

The same argument from the other chambers gives three more equations. The only difference is that leaving through the tunnel at B, C or D means E_A has *failed*, so that branch contributes $\frac{1}{3}\cdot 0$:

$$\begin{aligned} p_B &= \tfrac{1}{3}p_A + \tfrac{1}{3}p_C, \\ p_C &= \tfrac{1}{3}p_B + \tfrac{1}{3}p_D, \\ p_D &= \tfrac{1}{3}p_A + \tfrac{1}{3}p_C. \end{aligned}$$

The equations for p_B and p_D have the same right-hand side, so $p_B = p_D$; call this common value b. (This is the reflection symmetry in AC.) Then

$$p_A = \tfrac{1}{3} + \tfrac{2}{3}b, \qquad p_C = \tfrac{2}{3}b,$$

and substituting into the equation for p_B,

$$b = \tfrac{1}{3}\left(\tfrac{1}{3} + \tfrac{2}{3}b + \tfrac{2}{3}b\right) = \tfrac{1}{9} + \tfrac{4}{9}b \quad\Longrightarrow\quad \tfrac{5}{9}b = \tfrac{1}{9} \quad\Longrightarrow\quad b = \tfrac{1}{5}.$$

Hence

$$p_A = \tfrac{1}{3} + \tfrac{2}{15} = \frac{7}{15}, \qquad \text{and also} \qquad p_B = p_D = \tfrac{1}{5},\ \ p_C = \tfrac{2}{15}.$$

> **Ask yourself** How do we know the rabbit leaves at all, so that these equations really pin down the probabilities?

At every step the rabbit leaves with probability $\frac{1}{3}$, so the chance it is still underground after n steps is $(\frac{2}{3})^n \to 0$: it leaves with probability 1. The four linear equations also have only one solution (we solved them without any choice), so the values above are the probabilities.

Part (iii)

Let N be the event that the rabbit never visits C. We want

$$\mathrm{P}(N \mid E_A) = \frac{\mathrm{P}(N \cap E_A)}{\mathrm{P}(E_A)},$$

with the rabbit starting in A, so the denominator is $p_A = \frac{7}{15}$.

For the numerator, let q_A, q_B, q_D be the probabilities of leaving through the tunnel at A *without ever entering* C, given that the rabbit is currently in A, B, D respectively (and has not yet visited C). First-step analysis works exactly as before, except that a move into C now counts as failure:

$$q_A = \tfrac{1}{3} + \tfrac{1}{3}q_B + \tfrac{1}{3}q_D,$$
$$q_B = \tfrac{1}{3}q_A + \tfrac{1}{3}\cdot 0 + \tfrac{1}{3}\cdot 0,$$

since from B the rabbit either runs back to A, runs into C (failure) or leaves through the wrong tunnel (failure). By symmetry $q_D = q_B = \frac{1}{3}q_A$, so

$$q_A = \tfrac{1}{3} + \tfrac{2}{9}q_A \quad\Longrightarrow\quad \tfrac{7}{9}q_A = \tfrac{1}{3} \quad\Longrightarrow\quad q_A = \tfrac{3}{7}.$$

Therefore

$$\mathrm{P}(N \mid E_A) = \frac{3/7}{7/15} = \frac{45}{49}.$$

Checks

- The four values p_A, p_B, p_C, p_D are the probabilities of leaving from A starting in each chamber. By symmetry they are also the probabilities, starting from A, of leaving from A, B, C, D respectively, so they must add to 1: indeed $\frac{7}{15} + \frac{3}{15} + \frac{2}{15} + \frac{3}{15} = 1$.
- $p_A > p_B = p_D > p_C$, as it should be: the further the rabbit starts from A, the less likely it is to leave there.
- The answer to (iii) is close to 1: a rabbit that leaves from A usually did so after only a short trip. A simulation of 4×10^5 rabbits gave 0.469 for p_A and 0.918 for the conditional probability, against $\frac{7}{15} \approx 0.467$ and $\frac{45}{49} \approx 0.918$.

Common slips

- Including the $\frac{1}{3}$ "leave now" term in the equations for p_B, p_C, p_D: leaving from any tunnel other than A's contributes 0.
- Giving $q_A = \frac{3}{7}$ as the answer to (iii). That is $\mathrm{P}(N \cap E_A)$; the question asks for a conditional probability, so we still divide by p_A.

Beyond STEP

The rabbit is performing a *Markov chain*: its next move depends only on where it is now. The surface is an *absorbing state*, and p_A, p_B, p_C, p_D are *absorption probabilities*, which first-step analysis always turns into a system of linear equations. The same idea gives expected times to absorption, gambler's-ruin probabilities and much more. Grimmett and Welsh, *Probability: An Introduction*, has a good first treatment of Markov chains.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.